

The linear Paulsen bound: a streamlined proof

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Abstract

Every real ε -nearly equal-norm Parseval frame of n vectors in $\mathbb{R}^d$ is within squared Frobenius distance $C\varepsilon d$ of an equal-norm Parseval frame, where C is a universal constant. This is a reorganised and partly simplified account of a recent proof of this bound (produced with an AI system and formalised in Lean). The argument has one deterministic engine—a static row-scaling theorem driven by an ℓ_∞ Poisson estimate for the squared-Gram Laplacian—and two random “seeds” that make this engine applicable: a filtered Gaussian tangent when $d \gtrsim \log n$, and a row-tangent Gaussian when $n \gtrsim d^2$. Compared with the original manuscript, a sharper remainder estimate lets the many-row seed work at every error level below an absolute constant, which removes the random block partition and the d^6 block size from the global argument; the scaling lemmas are collected into one toolbox; and all probabilistic estimates are proved in full.

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1 The result and the plan

1.1 Statement

A frame of n vectors in $\mathbb{R}^d$ is a matrix $X \in \mathbb{R}^{n \times d}$ whose i th row is x_i^T . Distances are *squared* Frobenius distances $\|X - Y\|_F^2 = \sum_i \|x_i - y_i\|^2$, with no normalisation. Throughout,

$$a = d/n.$$

X is *Parseval* if $X^T X = I_d$ and *equal-norm* if $\|x_i\|^2 = a$ for all i ; an equal-norm Parseval frame is an *ENP frame*. X is an ε -*nearly ENP frame* if

$$(1 - \varepsilon)I_d \preceq X^T X \preceq (1 + \varepsilon)I_d, \quad (1 - \varepsilon)a \leq \|x_i\|^2 \leq (1 + \varepsilon)a \quad (1 \leq i \leq n). \quad (1.1)$$

Theorem 1.1 (Linear Paulsen bound). *There is an absolute constant C such that for all $1 \leq d \leq n$, all $0 < \varepsilon < 1$ and every real ε -nearly ENP frame $X \in \mathbb{R}^{n \times d}$ there is an ENP frame $W \in \mathbb{R}^{n \times d}$ with $\|X - W\|_F^2 \leq C\varepsilon d$.*

The bound is optimal up to the constant. Kwok, Lau, Lee and Ramachandran [3] proved the first bound independent of n , $O(\varepsilon d^{13/2})$; Hamilton and Moitra [2] improved it to $O(\varepsilon d^2)$; Lau and Ramachandran announced the linear bound [4]. The present text reorganises the proof of [5]; [Appendix B](#) lists what was changed.

For a Parseval U write $P = UU^T$ for the orthogonal projection onto its column space; then $P_{ii} = \|u_i\|^2$. The problem has an equivalent form in which the error is additive.

Theorem 1.2 (Projection form). *There is an absolute constant C such that for every $n \geq 1$ and every rank- d orthogonal projection P on $\mathbb{R}^n$, with $\beta = \max_i |P_{ii} - d/n|$, there is a rank- d orthogonal projection Q with $Q_{ii} = d/n$ for all i and $\|P - Q\|_F^2 \leq Cn\beta$.*

Both β and $\|P - Q\|_F$ are unchanged when (P, Q) is replaced by $(I - P, I - Q)$, so in [Theorem 1.2](#) one may assume $d \leq n/2$. The two theorems are derived from each other and from the main proposition in [Section 6](#).

Conventions. c, C denote positive absolute constants whose value may change between occurrences; constants with subscripts or names are fixed once chosen. $\|\cdot\|_{\text{op}}$ is the Euclidean operator norm, $\mathbf{1}$ the all-ones vector, $\Pi = I - n^{-1}\mathbf{1}\mathbf{1}^T$ the projection onto $\mathbf{1}^\perp$, $\text{Diag}(x)$ the diagonal matrix with diagonal x , and $\text{diag}(M)$ the diagonal of M as a vector. $A \preceq B$ is the Loewner order. A Laplacian inequality “on $\mathbf{1}^\perp$ ” means the corresponding quadratic forms are compared on vectors orthogonal to $\mathbf{1}$. All Gaussian vectors are real and centred.

1.2 The plan

The engine: scaling. Let U be Parseval with projection P and diagonal $p = \text{diag } P$. For $s \in \mathbb{R}^n$ the matrix $\text{Diag}(e^s)U$ has a column space with projection $P(s)$, and $s \mapsto \text{diag } P(s)$ is the gradient of the convex function $\frac{1}{2} \log \det(U^T \text{Diag}(e^{2s})U)$, whose Hessian at $s = 0$ is $2L_P$, where

$$L_P = \text{Diag}(p) - P \circ P$$

is the graph Laplacian with edge weights P_{ij}^2 (Lemma 2.2). To first order, then, changing the diagonal by $\delta \perp \mathbf{1}$ requires $s \approx \frac{1}{2}L_P^+ \delta$, and the cost of the move is about $s^T L_P s \approx \frac{1}{4} \delta^T L_P^+ \delta$. If L_P has spectral gap λ and $|\delta_i| \leq \varepsilon a$, the cost is at most $n\varepsilon^2 a^2 / \lambda$. For a “random-looking” P one has $\lambda \approx a$ and the cost is $\varepsilon^2 d$, but L_P can be arbitrarily badly connected: for an exact direct sum the components cannot exchange diagonal mass at all.

Perturb, then scale. Perturb U by a random horizontal tangent of size $t = \sqrt{K\varepsilon}$, at cost $O(t^2 d) = O(\varepsilon d)$. Generic entries of the new projection then have size $t\sqrt{a/n}$, so most edges acquire weight $\gtrsim at^2/n$ and the new Laplacian has gap $\gtrsim at^2$ —in fact an ℓ_∞ Poisson constant $O(1/(at^2))$ (Definition 2.6 and Lemma 2.11). The static balancing theorem (Theorem 2.8) then removes any diagonal error β with $\beta \cdot O(1/(at^2)) \leq \frac{1}{4}$ exactly, at cost $8n\beta = O(t^2 d)$. The ℓ_∞ (rather than spectral) formulation is essential: the scaling must stay in a box $\|s\|_\infty \leq 1$, and an ℓ_2 argument loses a factor $\sqrt{n}$.

The difficulty. So the perturbed frame must have diagonal error $\leq \theta at^2$ in ℓ_∞ for a small fixed θ . The initial error $\varepsilon a = at^2/K$ is fine. But a random tangent of size t moves each diagonal entry *linearly* by about $t\sqrt{a/n}$, which exceeds at^2 as soon as $\varepsilon \lesssim 1/d$. Two different constructions avoid this.

Seed 1: many rows ($n \geq Bd^2$, Section 4). Perturb each row *orthogonally to itself* and renormalise. Row norms stay exact; the cost is that the frame operator is no longer exactly I . Its first order change is removed by conditioning the Gaussian on a subspace of codimension $\leq d^2$, and its second-order fluctuation is an average over n rows, of size $t^2 d / \sqrt{n}$. The spectral error of the perturbed frame is therefore $\eta + O(t^2 d / \sqrt{n} + t^2 d^2 / n + t^3)$ with $t^2 = K\eta$, which is at most θt^2 once K and B are large and t is small; whitening turns a spectral error δ into a diagonal error at most $2a\delta$. (In [5] the remainder was bounded by $O(t^3 d)$, which forced $\eta \leq c/d^2$ and a separate block-partition argument; the sharper bound $O(t^3)$ uses the first-order constraint once more, see Lemma 4.4.)

Seed 2: moderate rows ($d \geq A \log 2n$, Section 5). Perturb in the horizontal tangent space of the Grassmannian (so the frame stays exactly Parseval) with a Gaussian whose covariance is filtered: $C_\rho = \rho(I - B)(\rho I + B)^{-1}$, where B is the normalised Gram operator of the linear diagonal

map. The filter makes the linear diagonal change $O(t\sqrt{\rho a \log n/n})$, negligible once $\rho \asymp t^2$ and $d \gg \log n$, while leaving weak (connecting) directions untouched. The price is a second-order bias $t^2 b_*$ which is not small in ℓ_∞ but is small in the dual Laplacian norm, $|\langle x, b_* \rangle|^2 \leq (2a/\rho)x^T L_P x$; it is cancelled exactly by one small scaling (Lemma 2.10). All other errors are $\leq \theta a t^2$ by concentration, which is where $d \geq A \log 2n$ is used.

Assembly (Section 6). After complementation and normalisation it suffices to treat equal-norm frames with $2d \leq n$ and spectral error η . If $d < D$ (a large absolute constant), the Hamilton–Moitra bound $\eta d(d-1) \leq D \eta d$ suffices (Section 3). If $d \geq D$ and $n \geq B d^2$, use Seed 1. If $d \geq D$ and $n < B d^2$, then $A \log 2n \leq A \log(2B d^2) \leq d$ and Seed 2 applies.

regime	construction	where
$d < D$	radial isotropic position, ordered coordinates (Hamilton–Moitra)	Section 3
$d \geq D, n \geq B d^2$	row-tangent Gaussian, row normalisation, whitening, balancing	Section 4
$d \geq D, 2d \leq n < B d^2$	filtered horizontal Gaussian, bias correction, balancing	Section 5
$d > n/2$	complement	Section 6

2 The scaling toolbox

Throughout this section $U \in \mathbb{R}^{n \times d}$ is Parseval, $P = U U^T$, $p = \text{diag } P$, and $L = L_P = \text{Diag}(p) - P \circ P$.

2.1 Projections, distances, and the squared-Gram Laplacian

Lemma 2.1 (Frame versus projection distance). *Let $U, W \in \mathbb{R}^{n \times d}$ be Parseval with projections P, Q , and let $\sigma_1, \dots, \sigma_d \in [0, 1]$ be the singular values of $U^T W$. Then*

$$\min_{R \in O(d)} \|U - WR\|_F^2 = 2 \sum_k (1 - \sigma_k) \leq \|P - Q\|_F^2 = 2 \sum_k (1 - \sigma_k^2) = 2 \|(I - P)W\|_F^2 \leq 2\|U - W\|_F^2.$$

Right multiplication by $R \in O(d)$ preserves Parsevalness and every row norm.

Proof. $\|U - WR\|_F^2 = 2d - 2 \text{tr}(U^T W R)$, maximised over R by the polar factor, giving $2d - 2 \sum_k \sigma_k$. Next $\|P - Q\|_F^2 = 2d - 2 \text{tr}(PQ) = 2d - 2\|U^T W\|_F^2$ and $\|(I - P)W\|_F^2 = d - \|U^T W\|_F^2$. Finally $1 - \sigma \leq 1 - \sigma^2 \leq 2(1 - \sigma)$ for $\sigma \in [0, 1]$. $\square$

Lemma 2.2 (Energy identity). *L is the Laplacian of the weighted graph with weights P_{ij}^2 ($i \neq j$), $L \succeq 0$, $L\mathbf{1} = 0$, $L_{I-P} = L_P$, and for $x \in \mathbb{R}^n$, $X = \text{Diag}(x)$,*

$$x^T L x = \frac{1}{2} \sum_{i,j} P_{ij}^2 (x_i - x_j)^2 = \|(I - P)XU\|_F^2 = \frac{1}{2} \|[X, P]\|_F^2.$$

Proof. $\sum_j P_{ij}^2 = (P^2)_{ii} = p_i$, so the rows of $\text{Diag}(p) - P \circ P$ sum to zero and its off-diagonal entries are $-P_{ij}^2$; this gives the first formula. Next $\|(I - P)XU\|_F^2 = \text{tr}(X P X (I - P)) = \sum_i x_i^2 p_i - \sum_{i,j} x_i x_j P_{ij}^2$, and $[X, P]_{ij} = (x_i - x_j) P_{ij}$. For $I - P$ the off-diagonal weights are again P_{ij}^2 . $\square$

Lemma 2.3 (Scaled projections). *For $w \in \mathbb{R}_{>0}^n$ put $D = \text{Diag}(w)$, $M = (U^T D^2 U)^{-1/2}$ and $W = D U M$. Then W is Parseval with column space $D \text{im } U$, $\|M\|_{\text{op}} \leq 1/\min_i w_i$, and*

$$\|W W^T - P\|_F^2 = 2\|(I - P)W\|_F^2 \leq \frac{2 w^T L w}{\min_i w_i^2}.$$

Proof. $W^T W = MU^T D^2 U M = I$ and $U^T D^2 U \succeq (\min w_i)^2 I$. By Lemma 2.1 and Lemma 2.2, $\|(I - P)W\|_F^2 \leq \|M\|_{\text{op}}^2 \|(I - P)DU\|_F^2 = \|M\|_{\text{op}}^2 w^T L w$. $\square$

2.2 The log-determinant potential

For $s \in \mathbb{R}^n$ let $D_s = \text{Diag}(e^s)$, let $P(s)$ be the projection onto $D_s \text{im } U$, and

$$F(s) = \frac{1}{2} \log \det(U^T D_s^2 U).$$

Lemma 2.4. F is smooth and convex, $F(s + c\mathbf{1}) = F(s) + cd$, and

$$\nabla F(s) = \text{diag } P(s), \quad \nabla^2 F(s) = 2L_{P(s)}.$$

Moreover, if $\|s\|_\infty \leq r$ then

$$e^{-4r} L_P \preceq L_{P(s)} \preceq e^{4r} L_P. \quad (2.1)$$

Proof. Write $G = U^T D_s^2 U = \sum_i e^{2s_i} u_i u_i^T$. Then $\partial_i F = e^{2s_i} u_i^T G^{-1} u_i = P(s)_{ii}$ because $P(s) = D_s U G^{-1} U^T D_s$, and differentiating once more gives $\partial_j \partial_i F = 2\delta_{ij} P(s)_{ii} - 2P(s)_{ij}^2$, i.e. $\nabla^2 F = 2L_{P(s)} \succeq 0$. For (2.1) let $W_s = D_s U M_s$ with $M_s = G^{-1/2}$, so $\|M_s\|_{\text{op}} \leq e^r$, $\|D_s\|_{\text{op}} \leq e^r$. Since $(I - P(s))D_s P = (I - P(s))W_s M_s^{-1} U^T = 0$,

$$(I - P(s)) \text{Diag}(x) W_s = (I - P(s)) D_s (I - P) \text{Diag}(x) U M_s,$$

and Lemma 2.2 for W_s gives $x^T L_{P(s)} x \leq e^{4r} x^T L_P x$. The lower bound is the upper bound applied to the Parseval frame W_s and the scaling $-s$, since $D_{-s} W_s = U M_s$ has column space $\text{im } U$. $\square$

Lemma 2.5 (Exact scaling inequalities). For $z \in \mathbb{R}_{>0}^n$ let r be the diagonal of the projection onto $\text{Diag}(\sqrt{z}) \text{im } U$. Then, coordinatewise,

$$Lz \leq z \circ (r - p), \quad L(z^{-1}) \leq -z^{-1} \circ (r - p).$$

Proof. Let $G = U^T \text{Diag}(z) U$. The matrix $(G - z_i I) G^{-1} (G - z_i I)$ is positive semidefinite; evaluating its quadratic form at u_i and using $u_i^T G u_i = \sum_j P_{ij}^2 z_j$, $\|u_i\|^2 = p_i$, $z_i u_i^T G^{-1} u_i = r_i$ gives $\sum_j P_{ij}^2 z_j - 2z_i p_i + z_i r_i \geq 0$, which is the first inequality. Let $V \in \mathbb{R}^{n \times (n-d)}$ be an isometry onto $(\text{im } U)^\perp$. The subspaces $\text{Diag}(\sqrt{z}) \text{im } U$ and $\text{Diag}(z^{-1/2}) \text{im } V$ are orthogonal complements, so the projection onto the latter has diagonal $\mathbf{1} - r$, while $V V^T = I - P$ has diagonal $\mathbf{1} - p$ and Laplacian L (Lemma 2.2). The first inequality for (V, z^{-1}) is the second inequality. $\square$

2.3 Static balancing

Definition 2.6. A weighted graph Laplacian L on n vertices has *Poisson constant* K if for every $f \perp \mathbf{1}$ there is g with $Lg = f$ and $\|g\|_\infty \leq K\|f\|_\infty$.

If L has a finite Poisson constant then the graph is connected (on each component the image of L sums to zero). The Poisson constant is the $\ell_\infty \rightarrow \ell_\infty$ norm of the pseudo-inverse modulo constants; it is the quantity that controls a scaling in the box $\|s\|_\infty \leq 1$.

Lemma 2.7 (Median barrier). Let L have Poisson constant K , let $\beta \geq 0$ with $2\beta K < 1$, let $z \in \mathbb{R}_{>0}^n$, let m be a median of z (so that $\{z \leq m\}$ and $\{z \geq m\}$ both have at least $n/2$ elements), and suppose

$$(Lz)_i \leq \beta z_i \quad \text{when } z_i > m, \quad (Lz^{-1})_i \leq \beta z_i^{-1} \quad \text{when } z_i < m.$$

Then $\max_i z_i / \min_i z_i \leq (1 - 2\beta K)^{-2}$.

Proof. Claim. If $|S| \geq n/2$, $y \geq 0$, $y \leq m$ on S and $(Ly)_i \leq \beta y_i$ for $i \notin S$, then $(1 - 2\beta K) \max y \leq m$. Indeed, $f = \mathbf{1} - \frac{n}{|S|} \mathbf{1}_S$ has $f \perp \mathbf{1}$ and $\|f\|_\infty \leq 1$; take $Lg = f$ with $\|g\|_\infty \leq K$ and $h = g + K\mathbf{1} \in [0, 2K]^n$, so $(Lh)_i = 1$ for $i \notin S$. With $M = \max y$, the vector $v = y - m\mathbf{1} - \beta Mh$ is ≤ 0 on S and satisfies $(Lv)_i \leq \beta y_i - \beta M \leq 0$ off S . If $\max v > 0$, it is attained at some $i \notin S$, and $(Lv)_i = \sum_j w_{ij}(v_i - v_j) \leq 0$ forces $v_j = v_i$ for all neighbours j ; by connectedness the maximum propagates to S , a contradiction. Hence $y \leq m + 2\beta KM$, and evaluating at the maximum proves the claim.

Apply the claim to $y = z$, $S = \{z \leq m\}$, and to $y = z^{-1}$, $S = \{z \geq m\}$ (with median m^{-1}). This gives $(1 - 2\beta K) \max z \leq m$ and $(1 - 2\beta K)m \leq \min z$. $\square$

Theorem 2.8 (Static balancing). *Suppose L_P has Poisson constant K , and $q \in \mathbb{R}^n$ satisfies $\sum_i q_i = d$, $\|q - p\|_\infty \leq \beta$ and $\beta K \leq \frac{1}{4}$. Then there is a Parseval W with $\text{diag}(WW^T) = q$ and*

$$\|U - W\|_F^2 \leq \|UU^T - WW^T\|_F^2 \leq 8\|q - p\|_1.$$

W is obtained from U by a row scaling with ratio of scales at most 2, whitening, and an orthogonal right factor.

Proof. Let $\Phi(s) = F(s) - \langle q, s \rangle$, so $\partial_i \Phi = P(s)_{ii} - q_i$, and let s^* minimise Φ over the cube $[0, 1]^n$. Write $r = \text{diag } P(s^*)$. Coordinatewise optimality gives

$$r_i \leq q_i \text{ if } s_i^* > 0, \quad r_i \geq q_i \text{ if } s_i^* < 1. \quad (2.2)$$

Let $z = e^{2s^*}$ with median m . If $z_i > m$ then $s_i^* > 0$, so by Lemma 2.5 and (2.2) $(Lz)_i \leq z_i(r_i - p_i) \leq z_i(q_i - p_i) \leq \beta z_i$; if $z_i < m$ then $s_i^* < 1$ and likewise $(Lz^{-1})_i \leq \beta z_i^{-1}$. By Lemma 2.7, $\max s^* - \min s^* \leq -\log(1 - 2\beta K) \leq \log 2 < 1$. Hence s^* does not meet both faces: either $s_i^* > 0$ for all i , and then $r \leq q$, or some $s_j^* = 0$, all $s_i^* < 1$, and then $r \geq q$. Since $\sum r_i = d = \sum q_i$, in both cases $r = q$.

Let $w = e^{s^*}$, $D = \text{Diag}(w)$, $m_0 = \min w$, $W_0 = DU(U^T D^2 U)^{-1/2}$, and $\delta = q - p$. Lemma 2.5 with $r = q$ gives $Lz \leq z \circ \delta$ for $z = w^2$, hence $z^T Lz \leq \sum_i z_i^2 \delta_i \leq 16m_0^4 \|\delta\|_1$, using $z_i \leq 4m_0^2$. Since $(z_i - z_j)^2 \geq 4m_0^2(w_i - w_j)^2$, Lemma 2.2 gives $w^T Lw \leq 4m_0^2 \|\delta\|_1$, and Lemma 2.3 gives $\|W_0 W_0^T - P\|_F^2 \leq 8\|\delta\|_1$. Finally take $W = W_0 R$ with R from Lemma 2.1. $\square$

Remark 2.9. No existence theorem for scalings, no flow, and no genericity is used: the box supplies a minimiser, the median barrier keeps it away from one face, and the trace identity $\sum_i r_i = d$ does the rest.

2.4 Local correction in the dual norm

The balancing theorem needs ℓ_∞ control of the target. The next lemma handles a different kind of target error: one that is small in the norm dual to a Laplacian with a spectral gap, although possibly large in ℓ_∞ .

Lemma 2.10 (Local correction). *Let W be Parseval with projection P_W and diagonal p_W . Let J be symmetric with $J \succeq \sigma I$ on $\mathbf{1}^\perp$ ($\sigma > 0$) and*

$$c_1 J \preceq L_{P_W} \preceq C_1 J \quad \text{on } \mathbf{1}^\perp.$$

Let $f \perp \mathbf{1}$ satisfy $\langle f, x \rangle^2 \leq E x^T J x$ for $x \perp \mathbf{1}$, and let $0 < R \leq r$ with

$$E < \gamma^2 \sigma R^2, \quad \gamma := 2c_1 e^{-4r}. \quad (2.3)$$

Then there is $s \perp \mathbf{1}$ with $\|s\|_\infty < R$, $s^T J s \leq E/\gamma^2$, such that the Parseval frame W' obtained from $\text{Diag}(e^s)W$ by whitening satisfies $\text{diag}(W'W'^T) = p_W + f$ and $L_{P_{W'}} \succeq c_1 e^{-4r} J$ on $\mathbf{1}^\perp$. After an orthogonal right factor,

$$\|W' - W\|_F^2 \leq \|P_{W'} - P_W\|_F^2 \leq 2C_1 e^{4r} E/\gamma^2.$$

Proof. Let F be the potential of W and $\Phi(s) = F(s) - F(0) - \langle p_W + f, s \rangle$; minimise Φ over the compact convex set $\Omega = \{s \perp \mathbf{1} : \|s\|_\infty \leq R\}$. For $s \in \Omega$ with $\|s\|_\infty = R$, Lemma 2.4 gives

$$\langle \nabla \Phi(s), s \rangle = \int_0^1 2s^T L_{P(us)} s \, du - \langle f, s \rangle \geq 2c_1 e^{-4r} s^T J s - \sqrt{E s^T J s} > 0,$$

because $s^T J s \geq \sigma \|s\|_2^2 \geq \sigma R^2 > E/\gamma^2$. So $-s$ is a feasible descent direction and the minimiser s lies in the relative interior of Ω . There $\nabla \Phi(s) \in \mathbb{R}\mathbf{1}$, and $\sum_i \nabla \Phi(s)_i = d - d - 0 = 0$, so $\text{diag } P(s) = p_W + f$. Pairing $\nabla \Phi(s) = 0$ with s as above gives $\gamma s^T J s \leq \sqrt{E s^T J s}$, i.e. $s^T J s \leq E/\gamma^2$, and then $\|s\|_\infty \leq \|s\|_2 < R$ by (2.3). The Laplacian bound is (2.1). For the distance, Lemma 2.3 with $w = e^s$, $\min w_i \geq e^{-r}$ and $|e^{s_i} - e^{s_j}| \leq e^r |s_i - s_j|$ gives

$$\|P_{W'} - P_W\|_F^2 \leq 2e^{2r} (e^s)^T L_{P_W} e^s \leq 2e^{4r} s^T L_{P_W} s \leq 2C_1 e^{4r} s^T J s,$$

and Lemma 2.1 supplies the right factor. $\square$

2.5 Poisson constants from a dense core

Lemma 2.11 (Dense core with exceptional vertices). *Let L be the Laplacian of nonnegative symmetric weights w_{ij} on $G \sqcup B$, $G \neq \emptyset$, $|B| = b$. Assume*

$$\sum_{j \in G} \min\{w_{ij}, w_{kj}\} \geq \rho > 0 \quad (i, k \in G), \quad \max_i \sum_j w_{ij} \leq D_{\max},$$

and, if $B \neq \emptyset$, that $C = L_{BB}$ is invertible with $\|C^{-1}\|_{\infty \rightarrow \infty} \leq R$. Then L has Poisson constant

$$K = \frac{1 + \min\{b, D_{\max} R\}}{\rho} + R$$

(with $R = b = 0$ if $B = \emptyset$).

Proof. Write $L = \begin{pmatrix} A & -E \\ -E^T & C \end{pmatrix}$ with $E = (w_{ij})_{i \in G, j \in B} \geq 0$. C is a symmetric Z -matrix with

$C\mathbf{1}_B = E^T \mathbf{1}_G \geq 0$; since it is invertible and positive semidefinite it is positive definite, and $C^{-1} \geq 0$ entrywise (if $Cx = y \geq 0$ then, with x_- the negative part, $0 \leq x_-^T y = x_-^T Cx \leq -x_-^T Cx_-$, so $x_- = 0$). Put $H = C^{-1}E^T \geq 0$ and $T = EC^{-1} = H^T$; then $H\mathbf{1}_G = \mathbf{1}_B$ and $\mathbf{1}_G^T T = \mathbf{1}_B^T$. Each row sum of T is at most b (entries of a column of T are bounded by its column sum 1) and at most $D_{\max}R$ (as $T\mathbf{1}_B \leq RE\mathbf{1}_B$).

The Schur complement $S = A - EC^{-1}E^T$ is the Laplacian on G with weights $\tilde{w}_{ik} = w_{ik} + (EC^{-1}E^T)_{ik} \geq w_{ik}$ (its rows sum to zero because $C^{-1}E^T \mathbf{1}_G = \mathbf{1}_B$). Let $f \perp \mathbf{1}$ with $\|f\|_\infty \leq 1$, and put $h = f_G + Tf_B$; then $\sum h = \sum f = 0$ and $\|h\|_\infty \leq \Phi := 1 + \min\{b, D_{\max}R\}$. The weights $\tilde{w}$ inherit the overlap bound, so the graph on G is connected and $Sy = h$ is solvable. If $i, k \in G$ maximise and minimise y ,

$$h_i - h_k = \sum_{j \in G} [\tilde{w}_{ij}(y_i - y_j) + \tilde{w}_{kj}(y_j - y_k)] \geq \rho(y_i - y_k),$$

so after adding a constant $\|y\|_\infty \leq \Phi/\rho$. Then $g_G = y$, $g_B = Hy + C^{-1}f_B$ solves $Lg = f$ (block substitution), and $\|g_B\|_\infty \leq \Phi/\rho + R$ because H is row-stochastic. $\square$

Corollary 2.12. Suppose $b \leq n/10$, $\gamma > 0$, and every $i \in G$ has at least $\frac{4}{5}n$ indices $j \neq i$ with $w_{ij} \geq \gamma/n$. Then the overlap hypothesis holds with $\rho = \gamma/2$. Moreover:

(a) if $L \succeq \lambda \Pi$ with $\lambda > 0$, then $R \leq 2b/\lambda$;

(b) if $w_{ij} = P_{ij}^2$ for a projection P , $a > 0$, $P_{ii} \geq a/2$ on B , $\sum_{i \in B} P_{ii} \leq \frac{1}{4}$, and $\max_i P_{ii} \leq C_2 a$, then $R \leq 4/a$ and $D_{\max} R \leq 4C_2$.

Proof. Two sets of at least $\frac{4}{5}n$ indices share at least $\frac{3}{5}n$, of which at least $\frac{1}{2}n$ lie in G , giving overlap $\geq \gamma/2$. (a) For x supported on B , $x^T C x = x^T L x \geq \lambda(\|x\|^2 - (\sum x_i)^2/n) \geq \lambda(1 - b/n)\|x\|^2$, so $|(C^{-1})_{ij}| \leq \|C^{-1}\|_{\text{op}} \leq (\lambda(1 - b/n))^{-1}$; sum over a row. (b) By $P_{ij}^2 \leq P_{ii}P_{jj}$, $(C\mathbf{1}_B)_i = P_{ii} - \sum_{j \in B} P_{ij}^2 \geq P_{ii}(1 - \sum_{j \in B} P_{jj}) \geq a/4$; so C is strictly diagonally dominant, $C^{-1} \geq 0$, and $C^{-1}\mathbf{1}_B \leq (4/a)\mathbf{1}_B$. Also $D_{\max} \leq \max_i P_{ii}$. $\square$

2.6 From a seed to an exact frame

The two constructions produce the following object.

Definition 2.13 (Seed). Let $\Theta \geq 2$ and $t > 0$. A (Θ, t) -seed for an $n \times d$ frame X is a Parseval $V \in \mathbb{R}^{n \times d}$ with

$$\|V - X\|_F^2 \leq \Theta t^2 d, \quad L_{VV^T} \text{ has Poisson constant } K \leq \frac{\Theta}{at^2}, \quad \max_i |(VV^T)_{ii} - a| \leq \frac{at^2}{4\Theta}.$$

Corollary 2.14. If X has a (Θ, t) -seed, there is an ENP frame W with $\|X - W\|_F^2 \leq 4\Theta t^2 d$.

Proof. Apply Theorem 2.8 to V with $q = a\mathbf{1}$: $\beta K \leq \frac{1}{4}$, and $\|V - W\|_F^2 \leq 8n \cdot \frac{at^2}{4\Theta} \leq 2t^2 d$. Then $\|X - W\|_F^2 \leq 2\|X - V\|_F^2 + 2\|V - W\|_F^2 \leq (2\Theta + 4)t^2 d \leq 4\Theta t^2 d$. $\square$

3 Bounded rank: the Hamilton–Moitra estimate

This section reproduces the radial-isotropic and ordered-coordinate argument of Hamilton and Moitra [2], with a direct compactness proof of the existence of the radial scaling (as in [5]). It is used for d below a large absolute constant, and it also shows that ENP frames exist.

Lemma 3.1. Let $1 \leq d \leq n$, let $X \in \mathbb{R}^{n \times d}$ have $\|x_i\|^2 = a$ for all i , and let $\eta = \|X^T X - I\|_{\text{op}}$. There is an ENP frame W with $\|X - W\|_F^2 \leq \eta d(d-1)$.

Proof. If $d = 1$ then $X^T X = na = 1$ and $W = X$ works. Let $d \geq 2$.

Step 1: full spark. Suppose first that every $d \times d$ minor $\mu_S = \det(X_S)^2$ ($S \subset [n]$, $|S| = d$) is nonzero. Consider $F(s) = \frac{1}{2} \log \det(X^T \text{Diag}(e^{2s})X) - a \sum_i s_i$. By Cauchy–Binet,

$$F(s) = \frac{1}{2} \log \sum_{|S|=d} \mu_S e^{T(s)_S}, \quad T(s)_S = 2 \left(\sum_{i \in S} s_i - a \sum_i s_i \right),$$

where $T : \mathbb{R}^n \rightarrow \mathbb{R}^{\binom{n}{d}}$ is linear and $\sum_S T(s)_S = 0$ because each i lies in a fraction $d/n = a$ of the d -sets. The function $\mathcal{E}(y) = \sum_S \mu_S e^{y_S}$ restricted to the subspace $\text{im } T$ has the nonempty sublevel set $\{\mathcal{E} \leq \mathcal{E}(0)\}$, on which $y_S \leq \log(\mathcal{E}(0)/\mu_S)$ for every S and hence, by $\sum_S y_S = 0$, also y_S is bounded below. This set is compact, so $\mathcal{E}|_{\text{im } T}$ attains its minimum at some $T(s^*)$, and s^* minimises F .

Let $D = \text{Diag}(e^{s^*})$, $M = (X^T D^2 X)^{-1/2}$ and $W = DXM$. As in [Lemma 2.4](#), $\partial_i F = \|w_i\|^2 - a$, so W is ENP, and since $w_i = e^{s_i^*} M x_i$ has norm $\sqrt{a}$,

$$w_i = \sqrt{a} \frac{M x_i}{\|M x_i\|}.$$

Step 2: ordered coordinates. Replacing X, W by XO, WO for a suitable $O \in O(d)$ changes neither the hypotheses nor the distance, and makes $M = \text{Diag}(\lambda_1, \dots, \lambda_d)$ with $0 < \lambda_1 \leq \dots \leq \lambda_d$. Then $w_{ij} = \lambda_j x_{ij} / \sqrt{c_i}$ with $c_i = a^{-1} \sum_\ell \lambda_\ell^2 x_{i\ell}^2$, so w_{ij} has the sign of x_{ij} and $w_{ij}^2 - x_{ij}^2 = x_{ij}^2 (\lambda_j^2 / c_i - 1)$ is nonpositive and then nonnegative as j increases. As $\sum_j (x_{ij}^2 - w_{ij}^2) = a - a = 0$, the prefix sums $\Delta_i(k) = \sum_{j \leq k} (x_{ij}^2 - w_{ij}^2)$ are nonnegative, vanish at $k = d$, and

$$\sum_j |x_{ij}^2 - w_{ij}^2| = 2 \max_k \Delta_i(k) \leq 2 \sum_{k=1}^{d-1} \Delta_i(k).$$

Since $W^T W = I$ and $(X^T X)_{jj} \leq 1 + \eta$, $\sum_i \Delta_i(k) = \sum_{j \leq k} ((X^T X)_{jj} - 1) \leq k\eta$. Equal signs give $(x_{ij} - w_{ij})^2 \leq |x_{ij}^2 - w_{ij}^2|$, and therefore

$$\|X - W\|_F^2 \leq 2\eta \sum_{k=1}^{d-1} k = \eta d(d-1).$$

Step 3: general X . Let G be a Vandermonde matrix with rows $(1, \theta_i, \dots, \theta_i^{d-1})$ for distinct θ_i ; all its maximal minors are nonzero. Each maximal minor of $X + hG$ is a polynomial in h with nonzero leading coefficient, so for all but finitely many h the matrix $X + hG$ is full spark; choose such $h_k \rightarrow 0$ and normalise each row of $X + h_k G$ to length $\sqrt{a}$ (a positive diagonal left factor, which preserves full spark). The resulting $X_k \rightarrow X$ have ENP frames W_k with $\|X_k - W_k\|_F^2 \leq \eta_k d(d-1)$, $\eta_k \rightarrow \eta$. The set of ENP frames is compact; a limit point W of (W_k) is ENP and satisfies the bound. $\square$

Corollary 3.2. *For all $1 \leq d \leq n$ an ENP frame exists. Consequently every equal-norm $X \in \mathbb{R}^{n \times d}$ is within $\|X - W\|_F^2 \leq 2\|X\|_F^2 + 2\|W\|_F^2 = 4d$ of an ENP frame W .*

Proof. Apply [Lemma 3.1](#) to $x_i = \sqrt{a} e_i$ for all i . $\square$

4 The many-row seed

Theorem 4.1. *There are absolute constants $B, c_*, C_* > 0$ such that the following holds. Let $2 \leq d$, $n \geq Bd^2$, let $X \in \mathbb{R}^{n \times d}$ have $\|x_i\|^2 = a$ for all i , and suppose $\eta := \|X^T X - I\|_{\text{op}}$ satisfies $0 < \eta \leq c_*$. Then there is an ENP frame W with $\|X - W\|_F^2 \leq C_* \eta d$.*

In [\[5\]](#) the same construction is analysed under the stronger hypothesis $\eta \leq c/d^2$; the improvement comes from [Lemma 4.4](#). Throughout, $S = X^T X$, so $\|S - I\|_{\text{op}} \leq \eta \leq \frac{1}{2}$, and $e_i = x_i / \sqrt{a}$.

4.1 The noise

In the Frobenius space $\mathbb{R}^{n \times d}$ consider the subspaces

$$\mathcal{E} = \{Z : \langle x_i, z_i \rangle = 0 \ \forall i\}, \quad \mathcal{F} = \{Z \in \mathcal{E} : X^T Z + Z^T X = 0\},$$

and let $m = \dim \mathcal{E} - \dim \mathcal{F} \leq \dim \text{Sym}_d \leq d^2$. For a standard Gaussian $g \in \mathbb{R}^{n \times d}$ put

$$Z_0 = n^{-1/2} \Pi_{\mathcal{E}} g, \quad Z = n^{-1/2} \Pi_{\mathcal{F}} g, \quad Z_{\perp} = Z_0 - Z = n^{-1/2} \Pi_{\mathcal{E} \ominus \mathcal{F}} g.$$

Z and $Z_{\perp}$ are independent. The rows $z_{0,i} = n^{-1/2}(I - e_i e_i^T)g_i$ of Z_0 are independent. Since $\text{Cov}(Z) \preceq \text{Cov}(Z_0)$, each row z_i of Z is Gaussian with covariance $\Sigma_i \preceq n^{-1}(I - e_i e_i^T)$, and $\text{Cov}(\text{vec } Z) \preceq I/n$. Moreover

$$\mathbb{E}\|Z\|_{\text{F}}^2 = \frac{n(d-1) - m}{n} \leq d, \quad \mathbb{E}\|Z - Z_0\|_{\text{F}}^2 = \frac{m}{n} \leq \frac{d^2}{n}. \quad (4.1)$$

Lemma 4.2 (Row moments). *Let $r_i = \|z_i\|^2/a$, $\mu_i = \mathbb{E}r_i$ and $\delta_i = 1 - \mu_i$. Then $1/d \leq \delta_i \leq 1$, $\sum_i a\delta_i = 1 + m/n \leq 2$, and*

$$\mathcal{A} := a \sum_i \mathbb{E}[(r_i - 1)^2(1 + r_i)^2] \leq C_{\mathcal{A}},$$

with $C_{\mathcal{A}}$ absolute.

Proof. $\tilde{\Sigma}_i = \Sigma_i/a \preceq d^{-1}I$ has trace $\mu_i \leq (d-1)/d$, so $\text{tr } \tilde{\Sigma}_i^2 \leq 1/d$ and $\text{tr } \tilde{\Sigma}_i^4 \leq d^{-3}$. For $\xi_i = r_i - \mu_i$ the Gaussian moment formulas give $\mathbb{E}\xi_i^2 = 2 \text{tr } \tilde{\Sigma}_i^2 \leq 2/d$ and $\mathbb{E}\xi_i^4 = 12(\text{tr } \tilde{\Sigma}_i^2)^2 + 48 \text{tr } \tilde{\Sigma}_i^4 \leq 60/d^2$. Next $\sum_i a\delta_i = d - \mathbb{E}\|Z\|_{\text{F}}^2 = 1 + m/n$ by (4.1). Finally $(r_i - 1)^2 \leq 2\xi_i^2 + 2\delta_i^2$ and $(1 + r_i)^2 \leq 8 + 2\xi_i^2$, so

$$\mathbb{E}(r_i - 1)^2(1 + r_i)^2 \leq 16\mathbb{E}\xi_i^2 + 4\mathbb{E}\xi_i^4 + 16\delta_i^2 + 4\delta_i^2\mathbb{E}\xi_i^2 \leq \frac{C}{d} + C\delta_i^2,$$

and $a \sum_i (C/d + C\delta_i^2) \leq C + Ca \sum_i \delta_i \leq 3C$. $\square$

4.2 Row normalisation and the exact second-order identity

For $0 < t \leq 1$ set

$$v_i = \frac{x_i + tz_i}{\sqrt{1 + t^2 r_i}}, \quad v_{0,i} = \frac{x_i + tz_{0,i}}{\sqrt{1 + t^2 r_{0,i}}}, \quad r_{0,i} = \|z_{0,i}\|^2/a.$$

Since $z_i \perp x_i$, $\|v_i\|^2 = \|v_{0,i}\|^2 = a$ exactly. The map $y \mapsto \sqrt{a}y/\|y\|$ is the metric projection onto the closed ball of radius $\sqrt{a}$ on $\{\|y\| \geq \sqrt{a}\}$, hence 1-Lipschitz there. Thus

$$\|V - X\|_{\text{F}} \leq t\|Z\|_{\text{F}}, \quad \|V - V_0\|_{\text{F}} \leq t\|Z - Z_0\|_{\text{F}}, \quad \|V\|_{\text{op}} \leq \|X + tZ\|_{\text{op}}, \quad \|V_0\|_{\text{op}} \leq \|X + tZ_0\|_{\text{op}}, \quad (4.2)$$

the last two because $V = \text{Diag}(\kappa)(X + tZ)$ with $0 < \kappa_i \leq 1$.

Expanding $1/(1 + t^2 r_i) = 1 - t^2 c_i$ with

$$c_i = \frac{r_i}{1 + t^2 r_i},$$

and using $X^T Z + Z^T X = 0$, one finds the exact identity

$$V^T V = S + t^2 Q(Z) + R_3 + R_4, \quad Q(Z) = Z^T Z - \sum_i r_i x_i x_i^T, \quad (4.3)$$

$$R_3 = -t^3 \sum_i c_i (x_i z_i^T + z_i x_i^T), \quad R_4 = -t^4 \sum_i c_i (z_i z_i^T - r_i x_i x_i^T).$$

Lemma 4.3 (Mean and fluctuation of Q). $\|\mathbb{E}Q(Z) - (I - S)\|_{\text{op}} \leq m/n$ and $\mathbb{E}\|Q(Z) - \mathbb{E}Q(Z)\|_{\text{F}}^2 \leq 8d^2/n$.

Proof. For Z_0 : $\mathbb{E}Z_0^T Z_0 = n^{-1} \sum_i (I - e_i e_i^T) = I - S/d$ and $\mathbb{E}r_{0,i} = (d-1)/d$, so $\mathbb{E}Q(Z_0) = I - S/d - (1 - 1/d)S = I - S$. As $Z_0 = Z + Z_\perp$ with independent centred summands and Q is quadratic, $\mathbb{E}Q(Z) = \mathbb{E}Q(Z_0) - \mathbb{E}Q(Z_\perp)$, and $\mathbb{E}Q(Z_\perp)$ is the difference of the positive semidefinite matrices $\mathbb{E}Z_\perp^T Z_\perp$ and $\sum_i \mathbb{E}(\|z_{\perp,i}\|^2/a) x_i x_i^T$, both of trace $\mathbb{E}\|Z_\perp\|_{\text{F}}^2 = m/n$; so its norm is at most m/n .

Entry (α, β) of Q is the quadratic form $\sum_i z_i^T B_{\alpha\beta,i} z_i$ with $B_{\alpha\beta,i} = \frac{1}{2}(E_{\alpha\beta} + E_{\beta\alpha}) - (e_i)_\alpha (e_i)_\beta I$. For a Gaussian vector with covariance $\Sigma \preceq I/n$ and symmetric coefficient matrix $\mathbf{B}$, the variance of the quadratic form is $2\|\Sigma^{1/2} \mathbf{B} \Sigma^{1/2}\|_{\text{F}}^2 \leq 2\|\mathbf{B}\|_{\text{F}}^2/n^2$. Since $\sum_{\alpha,\beta} \|B_{\alpha\beta,i}\|_{\text{F}}^2 \leq 2d^2 + 2d \leq 4d^2$ for each i , summing over α, β and the n blocks gives the claim. $\square$

Lemma 4.4 (Remainder). Let $\bar{c} = 1/(1+t^2)$, $\Gamma = \text{Diag}(c_i - \bar{c})$ and $\Lambda = \text{Diag}(c_i r_i - \bar{c})$. Then

$$\|R_3\|_{\text{op}} \leq 2t^3 \|X\|_{\text{op}} \|\Gamma Z\|_{\text{F}}, \quad \|R_4\|_{\text{op}} \leq t^4 (\|Z\|_{\text{op}}^2 + \|Z\|_{\text{op}} \|\Gamma Z\|_{\text{F}} + \|S\|_{\text{op}} + \|X\|_{\text{op}} \|\Lambda X\|_{\text{F}}),$$

and $\mathbb{E}\|\Gamma Z\|_{\text{F}}^2 \leq \mathcal{A}$, $\mathbb{E}\|\Lambda X\|_{\text{F}}^2 \leq 4\mathcal{A}$.

Proof. Because $\sum_i (x_i z_i^T + z_i x_i^T) = X^T Z + Z^T X = 0$, the constant part $\bar{c}$ of the weights drops out of R_3 : $R_3 = -t^3 (X^T \Gamma Z + Z^T \Gamma X)$, which gives the first bound. Next $\sum_i c_i z_i z_i^T = \bar{c} Z^T Z + Z^T \Gamma Z$ and $\sum_i c_i r_i x_i x_i^T = \bar{c} S + X^T \Lambda X$, with $\|Z^T \Gamma Z\|_{\text{op}} \leq \|Z\|_{\text{op}} \|\Gamma Z\|_{\text{F}}$ and similarly for Λ , and $\bar{c} \leq 1$. For the moments,

$$c_i - \bar{c} = \frac{r_i - 1}{(1+t^2 r_i)(1+t^2)}, \quad c_i r_i - \bar{c} = \frac{(r_i - 1)(1 + r_i + t^2 r_i)}{(1+t^2 r_i)(1+t^2)},$$

so $|c_i - \bar{c}| \leq |r_i - 1|$ and $|c_i r_i - \bar{c}| \leq 2|r_i - 1|(1 + r_i)$. Hence $\|\Gamma Z\|_{\text{F}}^2 = a \sum_i (c_i - \bar{c})^2 r_i \leq a \sum_i (r_i - 1)^2 (1 + r_i)^2$ and $\|\Lambda X\|_{\text{F}}^2 = a \sum_i (c_i r_i - \bar{c})^2 \leq 4a \sum_i (r_i - 1)^2 (1 + r_i)^2$; take expectations and use Lemma 4.2. $\square$

Remark 4.5. The crude bound $\|R_3\|_{\text{op}} \leq 2t^3 \sum_i c_i \|x_i\| \|z_i\| \approx 2t^3 d$ is what forces $\eta \lesssim d^{-2}$ in [5]. Centring the weights at $\bar{c}$, which is allowed by the first-order constraint, replaces the n terms of size a by fluctuations of relative size $d^{-1/2}$, whose ℓ_2 -sum over the rows is $O(1)$.

4.3 A dense reference graph

Lemma 4.6. There are absolute $h, t_0 \in (0, 1]$ and $c_1 > 0$ such that for $t \leq t_0$, with probability at least $1 - ne^{-c_1(n-1)}$, every i has at least $0.9(n-1)$ indices $j \neq i$ with

$$|\langle v_{0,i}, v_{0,j} \rangle| \geq ht\sqrt{a/n}.$$

Proof. Write $z_{0,j}/\sqrt{a} = d^{-1/2} \gamma_j$ with $\gamma_j \sim N(0, I - e_j e_j^T)$ independent. Fix i and condition on row i ; let $f = v_{0,i}/\sqrt{a}$, a unit vector. For $j \neq i$,

$$\frac{\langle v_{0,i}, v_{0,j} \rangle}{a} = \frac{\nu_j}{N_j}, \quad \nu_j = u_j + \frac{t}{\sqrt{d}} \langle f, \gamma_j \rangle, \quad u_j = \langle f, e_j \rangle, \quad N_j = (1 + \frac{t^2}{d} \|\gamma_j\|^2)^{1/2} \geq 1,$$

and the target inequality reads $|\nu_j|/N_j \geq ht/\sqrt{d}$. By Markov, $\mathbb{P}(N_j > 2) \leq t^2/3$. On $\{N_j \leq 2\}$ failure requires $|\nu_j| < 2ht/\sqrt{d}$. Here $\nu_j \sim N(u_j, t^2(1 - u_j^2)/d)$. If $|u_j| \leq \frac{1}{2}$ the variance is at least

$3t^2/(4d)$ and the Gaussian density bound gives probability at most $4h/\sqrt{3\pi/2} \leq 3h$. If $|u_j| > \frac{1}{2}$ and $2ht \leq \frac{1}{4}$, failure requires $\frac{t}{\sqrt{d}}|\langle f, \gamma_j \rangle| > \frac{1}{4}$, of probability at most $16t^2/d$. Choose h and t_0 so that $3h + t_0^2/3 + 8t_0^2 \leq \frac{1}{50}$. The failure indicators for $j \neq i$ are conditionally independent with probabilities at most $\frac{1}{50}$, so by the exponential Markov inequality (with parameter 1) more than $0.1(n-1)$ failures have probability at most $\exp(-(n-1)[0.1 - \frac{e-1}{50}]) \leq e^{-c_1(n-1)}$. Take a union bound over i . $\square$

4.4 Whitening

Lemma 4.7. *Let V be equal-norm with $\delta = \|V^T V - I\|_{\text{op}} \leq \frac{1}{2}$, $U = V(V^T V)^{-1/2}$ and $P = UU^T$. Then $|P_{ii} - a| \leq 2a\delta$, $\sum_j (P - VV^T)_{ij}^2 \leq 6a\delta^2$ for each i , and $\|U - V\|_{\text{F}}^2 \leq d\delta^2$.*

Proof. $P_{ii} = v_i^T (V^T V)^{-1} v_i \in [a/(1+\delta), a/(1-\delta)]$. With $E = (V^T V)^{-1} - I$, $\|E\|_{\text{op}} \leq \delta/(1-\delta) \leq 2\delta$, row i of $P - VV^T = VE V^T$ has squared norm $v_i^T E V^T V E v_i \leq 4\delta^2(1+\delta)a$. Finally the singular values σ_k of V satisfy $|\sigma_k - 1| \leq |\sigma_k^2 - 1| \leq \delta$. $\square$

4.5 Proof of Theorem 4.1

Constants. Fix h, t_0, c_1 from Lemma 4.6. Put $\kappa = h^2/200$ and $C_{\text{mix}} = 288/h^2 + 4$, then $\Theta = \max\{42, C_{\text{mix}}\}$ and $\theta = \min\{1/(8\Theta), h/60, 1/16\}$. Let $C_R = 300 + 100\sqrt{C_{\mathcal{A}}}$, choose $t_1 = \min\{t_0, \theta/(16C_R)\}$, and set

$$t^2 = 4\eta/\theta, \quad c_* = \theta t_1^2/4,$$

so that $\eta \leq c_*$ gives $t \leq t_1$. Finally choose B so large that $n \geq Bd^2$ implies $d^2/n \leq \theta/4$, $\sqrt{160d^2/n} \leq \theta/4$, $|B_0| \leq n/10$ and $\sum_{i \in B_0} 2a \leq \frac{1}{4}$ for the set B_0 below (which has $|B_0| \leq C'd/\kappa$), and $ne^{-c_1(n-1)} \leq 0.01$.

The event. By Markov's inequality, (4.1), Lemmas 4.3 and 4.4, the net bound (Lemma A.1(d)) for Z and Z_0 , and Lemma 4.6, with positive probability all of the following hold:

$$\|Z\|_{\text{F}}^2 \leq 20d, \quad \|Z - Z_0\|_{\text{F}}^2 \leq \frac{20d^2}{n}, \quad \|Q - \mathbb{E}Q\|_{\text{F}}^2 \leq \frac{160d^2}{n}, \quad \|\Gamma Z\|_{\text{F}}^2 \leq 20\mathcal{A}, \quad \|\Lambda X\|_{\text{F}}^2 \leq 80\mathcal{A},$$

$\|Z\|_{\text{op}}, \|Z_0\|_{\text{op}} \leq 16$, and the conclusion of Lemma 4.6. Fix such a sample.

Spectral error. By (4.3), $V^T V - I = (1-t^2)(S - I) + t^2(\mathbb{E}Q - (I - S)) + t^2(Q - \mathbb{E}Q) + R_3 + R_4$. Since $\|X\|_{\text{op}} \leq \sqrt{2}$, $\|S\|_{\text{op}} \leq \frac{3}{2}$ and $t \leq 1$, Lemma 4.4 and the event give $\|R_3\|_{\text{op}} + \|R_4\|_{\text{op}} \leq t^3(2\sqrt{40\mathcal{A}} + 256 + 16\sqrt{20\mathcal{A}} + \frac{3}{2} + \sqrt{160\mathcal{A}}) \leq C_R t^3$, and therefore

$$\delta := \|V^T V - I\|_{\text{op}} \leq \eta + t^2 \frac{d^2}{n} + t^2 \sqrt{\frac{160d^2}{n}} + C_R t^3 \leq \frac{\theta}{4} t^2 + \frac{\theta}{4} t^2 + \frac{\theta}{4} t^2 + \frac{\theta}{16} t^2 < \theta t^2.$$

Graph. By (4.2), $\|V\|_{\text{op}}, \|V_0\|_{\text{op}} \leq \sqrt{2} + 16 \leq 18$ and $\|VV^T - V_0V_0^T\|_{\text{F}} \leq 36t\|Z - Z_0\|_{\text{F}}$, so $\|VV^T - V_0V_0^T\|_{\text{F}}^2 \leq C't^2d^2/n$. Let B_0 be the set of rows whose contribution to this sum exceeds κat^2 ; then $|B_0| \leq C'd/\kappa$. A row $i \notin B_0$ has at most $\kappa at^2/((h/2)^2 t^2 a/n) = n/50$ indices with $|(VV^T - V_0V_0^T)_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$, so by Lemma 4.6 it has at least $0.85n$ indices $j \neq i$ with $|(VV^T)_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$.

Let $U = V(V^T V)^{-1/2}$, $P = UU^T$. By Lemma 4.7 a row loses at most $6a\theta^2 t^4/((h/4)^2 t^2 a/n) \leq n/20$ of these indices, so each $i \notin B_0$ has at least $0.8n$ indices $j \neq i$ with $P_{ij}^2 \geq (h^2/16)at^2/n$. Also

$a/2 \leq P_{ii} \leq 2a$ and $\sum_{i \in B_0} P_{ii} \leq \frac{1}{4}$. [Corollary 2.12\(b\)](#) with $\gamma = h^2 at^2/16$, $C_2 = 2$ shows that L_P has Poisson constant at most $9 \cdot 32/(h^2 at^2) + 4/a \leq C_{\text{mix}}/(at^2)$.

Seed. $\|U - X\|_F^2 \leq 2d\delta^2 + 2t^2\|Z\|_F^2 \leq 42t^2d$, the diagonal error is at most $2a\delta \leq 2\theta at^2 \leq at^2/(4\Theta)$, and the Poisson constant is at most $\Theta/(at^2)$. So U is a (Θ, t) -seed, and [Corollary 2.14](#) gives an ENP frame within $4\Theta t^2 d = (16\Theta/\theta)\eta d$ of X . $\square$

5 The moderate-row seed

Theorem 5.1. *There are absolute constants $A, C_m, \varepsilon_0 > 0$ such that the following holds. Let $U \in \mathbb{R}^{n \times d}$ be Parseval with $2d \leq n$, $d \geq A \log(2n)$, and $|||u_i||^2 - a| \leq \varepsilon a$ for all i , where $0 < \varepsilon \leq \varepsilon_0$. Then there is an ENP frame $\widehat{U}$ with $\|\widehat{U} - U\|_F^2 \leq C_m \varepsilon d$.*

Throughout this section $P = UU^T$, $p = \text{diag } P$, $D_p = \text{Diag}(p)$, $L = L_P$, and $\varepsilon_0 \leq \frac{1}{2}$, so $\frac{a}{2} \leq p_i \leq \frac{3a}{2} \leq \frac{3}{4}$. For a symmetric $M \in \mathbb{R}^{n \times n}$, $\mathcal{L}(M) = \sum_{i < j} M_{ij}^2 (e_i - e_j)(e_i - e_j)^T$ is the Laplacian with weights M_{ij}^2 ; thus $L = \mathcal{L}(P)$ and $x^T \mathcal{L}(M)x = \frac{1}{2} \|\text{Diag } x, M\|_F^2$. For nonnegative symmetric weights w , $\mathcal{L}[w]$ denotes the Laplacian with weights w_{ij} . We use $\|\mathcal{L}[w]\|_{\text{op}} \leq 2 \max_i \sum_j w_{ij}$.

5.1 The horizontal tangent space and the filtered covariance

Fix an isometry $V \in \mathbb{R}^{n \times (n-d)}$ onto $(\text{im } U)^\perp$, and let $\mathcal{Z} = \mathbb{R}^{(n-d) \times d}$ with the Frobenius inner product. For $Z \in \mathcal{Z}$ the matrix $H = VZ$ satisfies $U^T H = 0$; its rows are $h_i = Z^T v_i$. Define

$$A : \mathcal{Z} \rightarrow \mathbb{R}^n, \quad (AZ)_i = v_i^T Z u_i = (H U^T)_{ii}, \quad A^* x = N_x := V^T \text{Diag}(x) U.$$

Then $AA^* = L$, i.e. $\|N_x\|_F^2 = x^T L x$ ([Lemma 2.2](#)). Let

$$\bar{A} = D_p^{-1/2} A, \quad S = \bar{A} \bar{A}^* = D_p^{-1/2} L D_p^{-1/2}, \quad B = \bar{A}^* \bar{A}.$$

Since $L \succeq 0$ and $S = I - D_p^{-1/2} (P \circ P) D_p^{-1/2}$ with $P \circ P \succeq 0$ (Schur product theorem),

$$0 \preceq S \preceq I, \quad 0 \preceq B \preceq I, \quad \text{tr}(I - S) = \sum_i \frac{P_{ii}^2}{p_i} = d. \quad (5.1)$$

Fix an orthonormal eigenbasis of S , and for each of its members y with eigenvalue $\mu > 0$ put $f_y = D_p^{-1/2} y$ and $Z_y = \mu^{-1/2} \bar{A}^* y = \mu^{-1/2} N_{f_y}$. Since $B Z_y = \mu Z_y$ and $\langle Z_y, Z_{y'} \rangle = \langle y, S y' \rangle / \sqrt{\mu \mu'}$, the Z_y form an orthonormal eigenbasis of B on $(\ker B)^\perp = \text{im } \bar{A}^*$; and since $p_i \geq a/2$,

$$\|f_y\|_\infty \leq \sqrt{2/a}. \quad (5.2)$$

Definition 5.2 (Filtered covariance). For $\rho > 0$ let $C_\rho = \rho(I - B)(\rho I + B)^{-1}$ on $\mathcal{Z}$, and let $Z \sim N(0, C_\rho/n)$, $H = VZ$, $Y = H U^T + U H^T$.

Lemma 5.3. (a) $0 \preceq C_\rho \preceq I$; $C_\rho = I$ on $\ker B$.

(b) $\text{Cov}(AZ) = n^{-1} D_p^{1/2} \rho S (I - S) (\rho I + S)^{-1} D_p^{1/2} \preceq (\rho/n) D_p$.

(c) $I - C_\rho = B + R_\rho$ with $R_\rho = \sum_{\mu > 0} r_\mu Z_y \otimes Z_y$, $r_\mu = \mu(1 - \mu)/(\rho + \mu) \geq 0$, and

$$\sum_{\mu > 0} r_\mu \leq d, \quad \sum_{\mu > 0} \frac{r_\mu}{\sqrt{\mu}} \leq \frac{d}{2\sqrt{\rho}}, \quad \sum_{\mu > 0} \frac{r_\mu}{\mu} \leq \frac{d}{\rho}.$$

Proof. (a) The eigenvalues of C_ρ are $\rho(1 - \mu)/(\rho + \mu) \in [0, 1]$. (b) By singular value calculus $\bar{A}C_\rho\bar{A}^* = \rho S(I - S)(\rho I + S)^{-1}$, whose eigenvalues $\rho\mu(1 - \mu)/(\rho + \mu)$ are at most ρ . (c) On the μ -eigenspace, $1 - \rho(1 - \mu)/(\rho + \mu) = \mu + \mu(1 - \mu)/(\rho + \mu)$. The sums follow from $\mu/(\rho + \mu) \leq 1$, $\sqrt{\mu}/(\rho + \mu) \leq 1/(2\sqrt{\rho})$, $1/(\rho + \mu) \leq 1/\rho$, and $\sum_{\mu>0}(1 - \mu) \leq \text{tr}(I - S) = d$. $\square$

5.2 The second-order mean

For $Z \in \mathcal{Z}$ define the quadratic diagonal

$$q_i(Z) = \|h_i\|^2 - \|Zu_i\|^2 = \|Z^T v_i\|^2 - \|Zu_i\|^2.$$

For a covariance Σ on $\mathcal{Z}$ write $q(\Sigma) = \mathbb{E}q(G)$ with $G \sim N(0, \Sigma)$; $q(\cdot)$ is linear, and $q(\zeta \otimes \zeta) = q(\zeta)$.

Lemma 5.4 (Mean). $\mathbb{E}q(Z) = a\mathbf{1} - p - b_0 - b_*$, where

$$b_0 = \frac{1}{n}L(p^{-1}), \quad \|b_0\|_\infty \leq \frac{12\varepsilon}{n}, \quad b_* = \frac{1}{n} \sum_{\mu>0} r_\mu q(Z_y),$$

and $b_* \perp \mathbf{1}$ with

$$\langle x, b_* \rangle^2 \leq \frac{2a}{\rho} x^T Lx \quad (x \in \mathbb{R}^n). \quad (5.3)$$

Proof. For $\Sigma = I$: $\mathbb{E}\|Z^T v_i\|^2 = d\|v_i\|^2 = d(1 - p_i)$ and $\mathbb{E}\|Zu_i\|^2 = (n - d)p_i$, so $q(I/n) = a\mathbf{1} - p$. Next $B = \sum_j \zeta_j \otimes \zeta_j$ with $\zeta_j = \bar{A}^* e_j = v_j u_j^T / \sqrt{p_j}$, and

$$q_i(\zeta_j) = \frac{(I - P)_{ij}^2 p_j - (1 - p_j) P_{ij}^2}{p_j}, \quad \sum_j q_i(\zeta_j) = 1 - p_i - \sum_j \frac{P_{ij}^2}{p_j} + p_i = (Lp^{-1})_i,$$

using $\sum_j (I - P)_{ij}^2 = 1 - p_i$ and $\sum_j P_{ij}^2 = p_i$. Also $(Lp^{-1})_i = \sum_j P_{ij}^2 (p_i^{-1} - p_j^{-1})$ and $|p_i^{-1} - p_j^{-1}| \leq 2\varepsilon a / ((1 - \varepsilon)a)^2$, so $|(Lp^{-1})_i| \leq (1 + \varepsilon)a \cdot 8\varepsilon/a \leq 12\varepsilon$. With Lemma 5.3(c) this gives the formula for $\mathbb{E}q(Z) = q(C_\rho/n)$.

For (5.3) put $X = \text{Diag}(x)$, $D_x = V^T X V$, $A_x = U^T X U$. Then $\langle x, q(Z) \rangle = \langle Z, D_x Z - Z A_x \rangle$, and since X and $F = \text{Diag}(f)$ commute,

$$D_x N_f - N_f A_x = V^T [X(I - P)F - F P X] U = V^T [F(I - P)X - X P F] U = D_f N_x - N_x A_f. \quad (5.4)$$

For $Z = Z_y = \mu^{-1/2} N_f$, $f = f_y$, this gives $\langle x, q(Z_y) \rangle = \mu^{-1} \langle N_f, D_f N_x - N_x A_f \rangle$, and as $\|N_f\|_F^2 = \mu$, $\|N_x\|_F^2 = x^T Lx$, $\|D_f\|_{\text{op}}, \|A_f\|_{\text{op}} \leq \|f\|_\infty$,

$$|\langle x, q(Z_y) \rangle| \leq \frac{2\|f_y\|_\infty}{\sqrt{\mu}} \sqrt{x^T Lx} \leq 2\sqrt{\frac{2}{a\mu}} \sqrt{x^T Lx}.$$

Summing with weights r_μ/n and using Lemma 5.3(c), $|\langle x, b_* \rangle| \leq \frac{1}{n} \cdot \frac{d}{2\sqrt{\rho}} \cdot 2\sqrt{2/a} \sqrt{x^T Lx} = \sqrt{2a/\rho} \sqrt{x^T Lx}$. Taking $x = \mathbf{1}$ gives $b_* \perp \mathbf{1}$. $\square$

Remark 5.5. The bias b_* need not be small in ℓ_∞ : near a direct sum whose components are joined through a few rows it is concentrated there. What (5.3) says is that it can be removed by a scaling of energy $O(at^4/\rho)$ (Lemma 2.10), which is negligible once $\rho \asymp t^2$ with a large constant.

5.3 The expected graph

For a direction $\zeta \in \mathcal{Z}$ write $Y_\zeta = V\zeta U^T + U\zeta^T V^T$; then $\|Y_\zeta\|_F^2 = 2\|\zeta\|_F^2$, and for a covariance $\Sigma = \sum_k \lambda_k \psi_k \otimes \psi_k$, $\mathbb{E}_\Sigma \mathcal{L}(Y) = \sum_k \lambda_k \mathcal{L}(Y_{\psi_k})$.

Lemma 5.6. *On $\mathbf{1}^\perp$,*

$$\mathbb{E} \mathcal{L}(Y) \succeq \frac{a}{4} I - \left(\frac{2}{n} + \frac{8}{d} + \frac{8}{\rho} \right) L.$$

Proof. Unfiltered noise. For $Z_0 \sim N(0, I/n)$ and $i \neq j$, $Y_{ij} = v_i^T Z_0 u_j + v_j^T Z_0 u_i$ has variance $n^{-1} \|v_i u_j^T + v_j u_i^T\|_F^2 = n^{-1} (p_i + p_j - 2p_i p_j - 2P_{ij}^2)$. As $p \leq \frac{3}{4}$, $p_i + p_j - 2p_i p_j \geq \frac{1}{4}(p_i + p_j) \geq \frac{a}{4}$, so $\mathbb{E} \mathcal{L}(Y_{Z_0}) \succeq \frac{a}{4n} \mathcal{L}[\mathbf{1}\mathbf{1}^T] - \frac{2}{n} L = \frac{a}{4} \Pi - \frac{2}{n} L$.

A commutator bound. For $f \in \mathbb{R}^n$, $F = \text{Diag}(f)$, put $T_f = Y_{N_f} = (I - P)FP + PF(I - P) = FP + PF - 2PFP$. Expanding,

$$[X, T_f] = (I - 2P)F[X, P] + [X, P]F(I - 2P), \quad (5.5)$$

with $I - 2P$ orthogonal. Hence $x^T \mathcal{L}(T_f)x \leq 4\|f\|_\infty^2 x^T Lx$, and for $f = e_k$, since $E_k[X, P]$ and $[X, P]E_k$ are the k th row and column of $[X, P]$, $x^T \mathcal{L}(T_{e_k})x \leq 2 \sum_j P_{kj}^2 (x_k - x_j)^2$.

Base loss. $B = \sum_k \zeta_k \otimes \zeta_k$ with $\zeta_k = \bar{A}^* e_k = N_{e_k} / \sqrt{p_k}$, so $Y_{\zeta_k} = T_{e_k} / \sqrt{p_k}$ and $\sum_k x^T \mathcal{L}(Y_{\zeta_k})x \leq \frac{2}{a} \cdot 2 \sum_{k,j} P_{kj}^2 (x_k - x_j)^2 = \frac{8}{a} x^T Lx$. So the base part B/n of the removed covariance removes at most $\frac{8}{an} L = \frac{8}{d} L$.

Residual loss. By (5.2), $\mathcal{L}(Y_{Z_y}) = \mu^{-1} \mathcal{L}(T_{f_y}) \preceq \frac{8}{a\mu} L$, and by Lemma 5.3(c) the residual part R_ρ/n removes at most $\frac{8}{an} \sum_\mu \frac{r_\mu}{\mu} L \leq \frac{8}{\rho} L$. $\square$

5.4 A good sample

Lemma 5.7. *There are absolute $h \in (0, 1)$ and $b_{\max} \in \mathbb{N}$ and, for each $\eta' \in (0, \frac{1}{16}]$, a constant $A_{\eta'}$ such that the following holds for all $\rho > 0$, $0 < t \leq 1$, whenever $d \geq A_{\eta'} \log(2n)$. Put $J = L + at^2 \Pi$. With positive probability:*

$$(S1) \quad \|Z\|_{\text{op}} \leq 16, \max_i \|h_i\|^2 \leq 3a, \max_i \|Y_i\|^2 \leq 800a;$$

$$(S2) \quad \max_i |(AZ)_i| \leq 3\sqrt{\rho a \log(2n)/n} \text{ and } \max_i |q_i(Z) - \mathbb{E} q_i(Z)| \leq \eta' a;$$

$$(S3) \quad \|\mathcal{L}(Y) - \mathbb{E} \mathcal{L}(Y)\|_{\text{op}} \leq \eta' a;$$

$$(S4) \quad |x^T \mathcal{C}(Y)x| \leq \frac{\eta'}{t} x^T Jx \text{ for } x \perp \mathbf{1}, \text{ where } \mathcal{C}(Y) = \sum_{i < j} 2P_{ij} Y_{ij} (e_i - e_j)(e_i - e_j)^T;$$

$$(S5) \quad \text{there is a deterministic set } \mathfrak{B} \text{ (depending only on } U, \rho) \text{ with } |\mathfrak{B}| \leq b_{\max} \text{ such that every } i \notin \mathfrak{B} \text{ has at least } 0.95n \text{ indices } j \neq i \text{ with } |P_{ij} + tY_{ij}| \geq h t \sqrt{a/n}.$$

Proof. We show that each item fails with probability at most $\frac{1}{10}$. Note $C_\rho \preceq I$, so every linear functional $\langle Z, \Psi \rangle$ has variance at most $\|\Psi\|_F^2/n$; we use the tools of Appendix A.

(S1): The net bound (Lemma A.1(d)) with $\sigma^2 = 1/n$ gives $\mathbb{P}(\|Z\|_{\text{op}} > 16) \leq 2 \cdot 5^n e^{-8n}$. The vector $h_i = Z^T v_i$ has covariance $\preceq n^{-1} I_d$ and trace $\leq a$; Lemma A.1(b) with $u = 2a$ (so $\min\{u^2/\|\cdot\|_F^2, u/\|\cdot\|_{\text{op}}\} \geq \min\{4d, 2d\}$) and a union bound give $\max_i \|h_i\|^2 \leq 3a$ off probability $2ne^{-cd}$. Then $\|Y_i\| \leq \|h_i\| + \|Zu_i\| \leq \sqrt{3a} + 16\sqrt{3a/2}$.

(S2): $(AZ)_i$ is Gaussian with variance $\leq \rho p_i/n \leq 3\rho a/(2n)$ (Lemma 5.3(b)); a union bound gives the first claim for $n \geq 5$. The form $q_i(Z) = \langle Z, \mathbf{M}_i Z \rangle$, $\mathbf{M}_i Z = v_i v_i^T Z - Z u_i u_i^T$, has $\|\mathbf{M}_i\|_{\text{op}} \leq 1$

and $\|\mathbf{M}_i\|_F^2 \leq 2(d + np_i^2) \leq 7d$; after the covariance factor $(C_\rho/n)^{1/2}$ these become $\leq 1/n$ and $\leq \sqrt{7d}/n$, and Lemma A.1(b) with $u = \eta'a$ gives failure probability $2n \exp(-c\eta'^2 d)$.

(S3): $\text{Cov}(\text{vec } Y) \preceq \frac{2}{n}I$ (the map $Z \mapsto Y_Z$ is $\sqrt{2}$ times an isometry) and $\mathbb{E}\|Y_i\|^2 \leq 3a$ (for unfiltered noise $\mathbb{E}\|Y_i\|^2 = a(1 - p_i) + (1 - a)p_i$, and filtering only decreases it). Apply Lemma A.2.

(S4): Let $w_{ij} = J^{-1/2}(e_i - e_j)$, so $\|w_{ij}\|^2 \leq 2/(at^2)$. The matrix $tJ^{-1/2}\mathcal{C}(Y)J^{-1/2} = \sum_{i < j} tY_{ij} \cdot 2P_{ij}w_{ij}w_{ij}^T$ is a Gaussian series whose coefficients have covariance $\preceq \frac{2t^2}{n}I$, so by covariance domination (Lemma A.1(a)) its matrix variance is at most

$$\frac{2t^2}{n} \sum_{i < j} 4P_{ij}^2 \|w_{ij}\|^2 w_{ij}w_{ij}^T \preceq \frac{16}{an} J^{-1/2} L J^{-1/2} \preceq \frac{16}{d} I,$$

and Lemma A.1(a) bounds its norm by $C\sqrt{\log(20n)/d} \leq \eta'$ off probability $\frac{1}{10}$.

(S5): *Variance bookkeeping.* By the proof of Lemma 5.6, for $i \neq k$ with $P_{ik}^2 \leq a/16$ the unfiltered variance of Y_{ik} is at least $a/(8n)$; at most 24 indices k have $P_{ik}^2 > a/16$. Row i of T_{e_j} is $\delta_{ij}e_j^T P + P_{ij}e_j^T - 2P_{ij}e_j^T P$, of squared norm at most $3(\delta_{ij}p_j + P_{ij}^2 + 4P_{ij}^2 p_j)$, so the base part removes from row i total variance $\frac{1}{n} \sum_j \|e_i^T T_{e_j}\|^2 / p_j \leq \frac{6}{an} \cdot 5p_i \leq \frac{45}{n}$, which exceeds $\frac{a}{32n}$ at no more than $1440n/d$ positions. The residual part removes total variance $\frac{1}{n} \sum_\mu r_\mu \|Y_{Z_\mu}\|_F^2 \leq 2a$ from the whole matrix. Fix $\delta_0 = 1/6400$ and let $\mathfrak{B}$ be the set of rows from which it removes more than $\delta_0 a$; then $|\mathfrak{B}| \leq 2/\delta_0 =: b_{\max}$, and a row $i \notin \mathfrak{B}$ loses more than $\frac{a}{32n}$ to it at no more than $32\delta_0 n$ positions. So for $d \geq 10^6$ every $i \notin \mathfrak{B}$ has at least $0.99n$ indices $k \neq i$ with $\text{Var}(Y_{ik}) \geq \frac{a}{16n}$.

Small ball. Let $\phi: \mathbb{R} \rightarrow [0, 1]$ be C^1 , equal to 1 on $[-h, h]$, vanishing outside $[-2h, 2h]$, with $|\phi'| \leq 2/h$, and for $i \notin \mathfrak{B}$ let $F_i = \frac{1}{n} \sum_{k \neq i} \phi((P_{ik} + tY_{ik})/(t\sqrt{a/n}))$. By the density bound for a Gaussian of variance $\geq \frac{at^2}{16n}$, $\mathbb{E}F_i \leq 0.01 + \frac{4h}{\sqrt{2\pi/16}} \leq 0.01 + 7h$. As a function of the row $(Y_{ik})_k$, F_i has gradient norm at most $\frac{1}{n} \cdot \frac{2}{h} \cdot \sqrt{n/a} \cdot \sqrt{n} = 2/(h\sqrt{a})$; the row depends on Z through a linear map of norm ≤ 2 , and $Z = (C_\rho/n)^{1/2}g$ with g standard. So F_i is $4/(h\sqrt{d})$ -Lipschitz in g , and Lemma A.1(c) gives $\mathbb{P}(F_i > \mathbb{E}F_i + 0.01) \leq 2\exp(-ch^2d)$. Take $h = 1/500$, so that $\mathbb{E}F_i + 0.01 \leq 0.04$, and a union bound over $i \notin \mathfrak{B}$. On the complement, at most $0.04n$ indices $k \neq i$ have $|P_{ik} + tY_{ik}| < ht\sqrt{a/n}$, so at least $n - 1 - 0.04n \geq 0.95n$ have the reverse inequality.

Finally choose $A_{\eta'}$ so large that $d \geq A_{\eta'} \log(2n)$ implies $d \geq 10^6$ and all the failure probabilities above are at most $\frac{1}{10}$ (there are five items, so their union has probability < 1). $\square$

5.5 Retraction

Fix a sample as in Lemma 5.7 and set

$$G = I + t^2 Z^T Z, \quad U_t = (U + tH)G^{-1/2}, \quad P_t = U_t U_t^T.$$

Since $U^T H = 0$ and $H^T H = Z^T Z$, $(U + tH)^T (U + tH) = G$ and U_t is Parseval.

Lemma 5.8. *There is an absolute C_E such that for $0 < t \leq 1$:*

- (a) $\|U_t - U\|_F^2 \leq C_E t^2 d$;
- (b) every row of $\mathbf{E} = P_t - P - tY$ satisfies $\sum_j \mathbf{E}_{ij}^2 \leq C_E a t^4$;
- (c) $\text{diag } P_t = p + 2tAZ + t^2 \mathbb{E}q(Z) + t^2(q(Z) - \mathbb{E}q(Z)) + \mathbf{r}$ with $\|\mathbf{r}\|_\infty \leq C_E a t^3$.

Proof. By (S1), $\|G^{-1} - I\|_{\text{op}}, \|G^{-1/2} - I\|_{\text{op}} \leq 256t^2$, $\|U + tH\|_{\text{op}} \leq 17$, $\|U + tH\|_F^2 \leq d(1 + 256t^2)$ and $\|u_i + th_i\| \leq 3\sqrt{a}$. (a) $\|U_t - U\|_F \leq \|U + tH\|_F \|G^{-1/2} - I\|_{\text{op}} + t\|Z\|_F \leq Ct\sqrt{d}$. (b)

$\mathbf{E} = (U + tH)(G^{-1} - I)(U + tH)^T + t^2 H H^T$, and row i has norm at most $3\sqrt{a} \cdot 256t^2 \cdot 17 + t^2 \sqrt{3a} \cdot 16$.
(c) Insert $G^{-1} = I - t^2 Z^T Z + t^4 (Z^T Z)^2 G^{-1}$ into $(P_t)_{ii} = (u_i + th_i)^T G^{-1} (u_i + th_i)$ and use $u_i^T h_i = (AZ)_i$:

$$(P_t)_{ii} = p_i + 2t(AZ)_i + t^2 q_i(Z) - 2t^3 u_i^T Z^T Z h_i - t^4 \|Zh_i\|^2 + t^4 (u_i + th_i)^T (Z^T Z)^2 G^{-1} (u_i + th_i),$$

and the last three terms are $O(at^3)$ by (S1). $\square$

Lemma 5.9 (Regularised seed). *There are absolute $t_* \in (0, 1]$ and C_J such that if $\eta' \leq \frac{1}{16}$, $\rho = D^2 t^2$ with $D \geq 8$, and $0 < t \leq t_*$, then on $\mathbf{1}^\perp$*

$$\frac{1}{32}J \preceq L_{P_t} \preceq C_J J,$$

and every $i \notin \mathfrak{B}$ has at least $0.9n$ indices $j \neq i$ with $|(P_t)_{ij}| \geq \frac{h}{2}t\sqrt{a/n}$.

Proof. Entrywise $(P + tY)_{ij}^2 = P_{ij}^2 + 2tP_{ij}Y_{ij} + t^2 Y_{ij}^2$, so $\mathcal{L}(P + tY) = L + t\mathcal{C}(Y) + t^2 \mathcal{L}(Y)$. On $\mathbf{1}^\perp$, by (S4), (S3) and Lemma 5.6,

$$\mathcal{L}(P + tY) \succeq L - \eta' J + t^2 \left(\frac{a}{4} - \eta' a \right) I - t^2 \left(\frac{2}{n} + \frac{8}{d} + \frac{8}{D^2 t^2} \right) L \succeq \left(\frac{7}{8} - 10t^2 - \frac{1}{8} \right) L + \frac{at^2}{8} I \succeq \frac{1}{8} J$$

for $t^2 \leq 1/40$. Also $\mathcal{L}(P + tY) \preceq 2L + 2t^2 \mathcal{L}(Y) \preceq 2L + 3200at^2 I$ by (S1). Now $(P_t)_{ij}^2 \geq \frac{1}{2}(P + tY)_{ij}^2 - \mathbf{E}_{ij}^2$ and $(P_t)_{ij}^2 \leq 2(P + tY)_{ij}^2 + 2\mathbf{E}_{ij}^2$, while $\mathcal{L}(\mathbf{E}) \preceq 2C_E at^4 I$ by Lemma 5.8(b). Hence $L_{P_t} \succeq \frac{1}{16}J - 2C_E at^4 I \succeq \frac{1}{32}J$ when $t^2 \leq 1/(64C_E)$, and $L_{P_t} \preceq (6400 + 4C_E)J$. Finally, in a row $i \notin \mathfrak{B}$ at most $C_E at^4 / (\frac{h^2}{4} t^2 \frac{a}{n}) = 4C_E t^2 n / h^2 \leq 0.05n$ indices have $|\mathbf{E}_{ij}| \geq \frac{h}{2}t\sqrt{a/n}$ once $t^2 \leq h^2/(80C_E)$; combine with (S5). $\square$

5.6 Removing the bias and balancing

Proof of Theorem 5.1. Choice of constants. Fix $h, b_{\max}$ (Lemma 5.7) and t_*, C_J, C_E (Lemmas 5.8 and 5.9); these are absolute. Put $C' = 1024e^3 C_J$ and $D = \max\{250, 40\sqrt{C'}/h\}$, which fixes $\rho = D^2 t^2$. Put

$$C_{\text{core}} = \frac{32(1 + b_{\max})}{h^2} + 64e b_{\max}, \quad C_\Sigma = 2C_E + 2C', \quad \Theta = \max\{2, C_{\text{core}}, C_\Sigma\},$$

let $K_0 = 20\Theta$ and $\eta' = 1/(20\Theta)$, and choose $A \geq A_{\eta'}$ so large that $d \geq A \log 2n$ implies $6D\sqrt{\log(2n)/d} \leq 1/(20\Theta)$ and $b_{\max} \leq n/10$. Finally choose $\varepsilon_0 \leq \frac{1}{2}$ so that $t = \sqrt{K_0\varepsilon}$ satisfies $t \leq t_*$, $C_E t \leq 1/(20\Theta)$ and $12\varepsilon \leq 1/(20\Theta)$ for $\varepsilon \leq \varepsilon_0$. The order of choices is acyclic, and none depends on n, d, ε or U . Take a sample as in Lemma 5.7.

Bias correction. Apply Lemma 2.10 to $W = U_t$ with J as above, $\sigma = at^2$, $c_1 = \frac{1}{32}$, $C_1 = C_J$, $f = t^2 b_*$ and $r = R = \frac{1}{4}$. By (5.3) and $L \preceq J$, $\langle f, x \rangle^2 \leq \frac{2at^4}{\rho} x^T J x$, so $E = 2at^2/D^2$; with $\gamma = 2c_1 e^{-1} = 1/(16e)$, the condition $E < \gamma^2 \sigma R^2 = at^2/(4096e^2)$ holds as $D \geq 250$. We obtain a Parseval U_* with

$$\text{diag } P_* = \text{diag } P_t + t^2 b_*, \quad \|U_* - U_t\|_F^2, \|P_* - P_t\|_F^2 \leq C' at^2/D^2, \quad L_{P_*} \succeq \frac{at^2}{32e} \Pi \text{ on } \mathbf{1}^\perp,$$

using $2C_1 e^{4r} E/\gamma^2 = 2C_J e \cdot \frac{2at^2}{D^2} \cdot 256e^2 = C' at^2/D^2$. Since $\|P_* - P_t\|_F^2 \leq C' at^2/D^2$, each row loses at most $C'(at^2/D^2)/(\frac{h^2}{16} t^2 \frac{a}{n}) \leq 0.1n$ of the indices of Lemma 5.9 when $D \geq 40\sqrt{C'}/h$; so every $i \notin \mathfrak{B}$ has at least $0.8n$ indices $j \neq i$ with $(P_*)_{ij}^2 \geq \frac{h^2}{16} \frac{at^2}{n}$.

Poisson constant. Corollary 2.12(a) with $G = \mathfrak{B}^c$, $b \leq b_{\max} \leq n/10$, $\gamma = h^2 at^2/16$ and $\lambda = at^2/(32e)$ gives $R \leq 64e b_{\max}/(at^2)$ and Poisson constant $K \leq (1+b_{\max})\frac{32}{h^2 at^2} + \frac{64eb_{\max}}{at^2} \leq \frac{C_{\text{core}}}{at^2}$.

Diagonal error. By Lemmas 5.4 and 5.8(c), the bias cancels exactly:

$$\text{diag } P_* - a\mathbf{1} = (1 - t^2)(p - a\mathbf{1}) + 2tAZ - t^2b_0 + t^2(q(Z) - \mathbb{E}q(Z)) + r.$$

By (S2) and $\rho = D^2 t^2$,

$$\frac{\|\text{diag } P_* - a\mathbf{1}\|_\infty}{at^2} \leq \frac{1}{K_0} + 6D\sqrt{\frac{\log 2n}{d}} + \frac{12\varepsilon}{d} + \eta' + C_E t \leq \frac{5}{20\Theta} = \frac{1}{4\Theta}.$$

Seed and cost. By Lemma 5.8(a) and the bias correction, $\|U_* - U\|_F^2 \leq 2C_E t^2 d + 2C' at^2/D^2 \leq C_\Sigma t^2 d$ with $C_\Sigma = 2C_E + 2C'$. So U_* is a (Θ, t) -seed for U , and Corollary 2.14 gives an ENP frame $\hat{U}$ with $\|\hat{U} - U\|_F^2 \leq 4\Theta t^2 d = 4\Theta K_0 \varepsilon d$. $\square$

6 Assembly

We use the two seeds through the following equal-norm formulation.

Proposition 6.1. *There is an absolute constant C_0 such that every $X \in \mathbb{R}^{n \times d}$ with $1 \leq d$, $2d \leq n$ and $\|x_i\|^2 = a$ for all i admits an ENP frame W with*

$$\|X - W\|_F^2 \leq C_0 d \|X^T X - I\|_{\text{op}}.$$

Proof. Let $\eta = \|X^T X - I\|_{\text{op}}$; if $\eta = 0$ take $W = X$. Let A, ε_0, C_m be the constants of Theorem 5.1 and B, c_*, C_* those of Theorem 4.1, and fix an integer $D \geq 2$ with $A \log(2Bd^2) \leq d$ for all $d \geq D$.

Case $d < D$. Lemma 3.1 gives $\|X - W\|_F^2 \leq \eta d(d-1) \leq D\eta d$.

Case $d \geq D$, $n \geq Bd^2$. If $\eta \leq c_*$ use Theorem 4.1; otherwise Corollary 3.2 gives cost $4d \leq (4/c_*)\eta d$.

Case $d \geq D$, $n < Bd^2$. Then $A \log(2n) \leq d$. If $\eta \leq \varepsilon_0/4$, let $U = X(X^T X)^{-1/2}$. The singular values of X lie in $[\sqrt{1-\eta}, \sqrt{1+\eta}]$, so $\|X - U\|_F^2 \leq d\eta^2$, and $\|u_i\|^2 = x_i^T (X^T X)^{-1} x_i \in [a/(1+\eta), a/(1-\eta)]$, so $|\|u_i\|^2 - a| \leq 2\eta a$. Theorem 5.1 with $\varepsilon = 2\eta$ gives an ENP W with $\|U - W\|_F^2 \leq 2C_m \eta d$, and $\|X - W\|_F^2 \leq 2d\eta^2 + 4C_m \eta d$. If $\eta > \varepsilon_0/4$ use the trivial cost $4d$. $\square$

Proof of Theorem 1.2. Replacing (P, Q) by $(I - P, I - Q)$ preserves β , ranks of complements and distances, so we may assume $d \leq n/2$; the case $d = 0$ is trivial. Write $P = UU^T$ with $U^T U = I$, $p_i = \|u_i\|^2$ and $\delta = \beta/a$. If $\delta \geq \frac{1}{2}$, any ENP W (Corollary 3.2) gives $\|P - WW^T\|_F^2 \leq 2d \leq 4n\beta$. Otherwise let $x_i = \sqrt{a} u_i / \|u_i\|$. Then $\|X - U\|_F^2 = \sum_i (\sqrt{p_i} - \sqrt{a})^2 \leq \sum_i (p_i - a)^2/a \leq \delta^2 d$ and $X^T X = U^T \text{Diag}(a/p)U$, so $\|X^T X - I\|_{\text{op}} \leq \max_i |a/p_i - 1| \leq \delta/(1-\delta) \leq 2\delta$. Proposition 6.1 gives an ENP W with $\|U - W\|_F^2 \leq 2\delta^2 d + 4C_0 \delta d \leq (1 + 4C_0)n\beta$, using $\delta d = n\beta$. Take $Q = WW^T$ and use Lemma 2.1. $\square$

Proof of Theorem 1.1. If $\varepsilon \geq \frac{1}{2}$, an ENP W gives $\|X - W\|_F^2 \leq 2\|X\|_F^2 + 2d \leq 6d \leq 12\varepsilon d$, since $\|X\|_F^2 = \text{tr } X^T X \leq (1 + \varepsilon)d$. Let $\varepsilon < \frac{1}{2}$ and $U = X(X^T X)^{-1/2}$, $P = UU^T$. As above $\|X - U\|_F^2 \leq \varepsilon^2 d$ and $P_{ii} = x_i^T (X^T X)^{-1} x_i \in [\frac{1-\varepsilon}{1+\varepsilon}a, \frac{1+\varepsilon}{1-\varepsilon}a]$, so $\beta = \max_i |P_{ii} - a| \leq 4\varepsilon a$. Theorem 1.2 gives Q with $\|P - Q\|_F^2 \leq 4C\varepsilon d$, and by Lemma 2.1 there is an isometry W spanning Q —hence ENP—with $\|U - W\|_F^2 \leq \|P - Q\|_F^2$. Then $\|X - W\|_F^2 \leq 2\varepsilon^2 d + 8C\varepsilon d$. $\square$

A Gaussian tools

Lemma A.1. *The following hold with absolute constants.*

(a) (Matrix series.) *Let $X = \sum_k g_k A_k$ with independent standard Gaussians g_k and symmetric $r \times r$ matrices A_k , and $v = \|\sum_k A_k^2\|_{\text{op}} = \|\mathbb{E}X^2\|_{\text{op}}$. Then $\mathbb{E}\|X\|_{\text{op}} \leq C\sqrt{v \log(2r)}$ and $\mathbb{P}\{\|X\|_{\text{op}} > C\sqrt{v \log(2r/\delta)}\} \leq \delta$ for $0 < \delta \leq 1$. If instead $X = \sum_\alpha \xi_\alpha M_\alpha$ with (ξ_α) a centred Gaussian vector of covariance $\preceq \gamma I$, then X has this form with $\sum_k A_k^2 = \mathbb{E}X^2 \preceq \gamma \sum_\alpha M_\alpha^2$.*

(b) (Quadratic forms.) *For g standard Gaussian and A symmetric, $\text{Var}(g^T A g) = 2\|A\|_{\text{F}}^2$ and*

$$\mathbb{P}\{|g^T A g - \text{tr } A| > u\} \leq 2 \exp\left[-\frac{1}{8} \min\left\{\frac{u^2}{\|A\|_{\text{F}}^2}, \frac{u}{\|A\|_{\text{op}}}\right\}\right].$$

For a Gaussian vector $\zeta = \Sigma^{1/2}g$, apply this to $\Sigma^{1/2}A\Sigma^{1/2}$, whose norms are at most $\|\Sigma\|_{\text{op}}$ times those of A .

(c) (Lipschitz functions.) *If F is C^1 with $\|\nabla F\| \leq \ell$ and g is standard Gaussian, then $\mathbb{P}\{|F(g) - \mathbb{E}F(g)| > u\} \leq 2 \exp(-2u^2/(\pi^2 \ell^2))$.*

(d) (Nets.) *If Z is an $r \times s$ centred Gaussian matrix with $\text{Cov}(\text{vec } Z) \preceq \sigma^2 I$, then $\mathbb{P}\{\|Z\|_{\text{op}} > u\} \leq 2 \cdot 5^{r+s} \exp(-u^2/(32\sigma^2))$.*

Proof. (a) Let $M_p = \mathbb{E} \text{tr } X^{2p}$. Gaussian integration by parts gives $M_p = \sum_k \sum_{j=0}^{2p-2} \mathbb{E} \text{tr}(A_k X^j A_k X^{2p-2-j})$. For symmetric A , even q and $0 \leq j \leq q$, diagonalising $X = \sum_l \lambda_l e_l e_l^T$ and using weighted AM–GM, $|\text{tr}(A X^j A X^{q-j})| = |\sum_{l,m} A_{lm}^2 \lambda_m^j \lambda_l^{q-j}| \leq \sum_{l,m} A_{lm}^2 |\lambda_l|^q = \text{tr}(A^2 |X|^q)$. Hence $M_p \leq (2p-1)vM_{p-1}$ and $M_p \leq r(2p-1)!! v^p \leq r(2pv)^p$. Markov's inequality for $\text{tr } X^{2p} \geq \|X\|_{\text{op}}^{2p}$ with $p = \lceil \log(r/\delta) \rceil$ and $u^2 = 2e^2 pv$ gives the tail; $p = \lceil \log 2r \rceil$ and Jensen give the mean. For the second statement write $\xi = \Gamma^{1/2}g$; then $X = \sum_k g_k A_k$ with $A_k = \sum_\alpha (\Gamma^{1/2})_{\alpha k} M_\alpha$, and for a vector w , $w^T \mathbb{E}X^2 w = \mathbb{E}\|Xw\|^2 = \text{tr}(\Gamma G_w) \leq \gamma \text{tr } G_w = \gamma \sum_\alpha \|M_\alpha w\|^2$, where G_w is the Gram matrix of $(M_\alpha w)_\alpha$.

(b) Diagonalise A with eigenvalues λ_j . For $|s| \leq 1/(4\|A\|_{\text{op}})$, $\log \mathbb{E}e^{s(g^T A g - \text{tr } A)} = \sum_j [-s\lambda_j - \frac{1}{2} \log(1 - 2s\lambda_j)] \leq 2s^2\|A\|_{\text{F}}^2$, since $-x - \frac{1}{2} \log(1 - 2x) = \sum_{k \geq 2} (2x)^k/(2k) \leq 2x^2$ for $|x| \leq \frac{1}{4}$. The exponential Markov inequality with $s = \min\{u/(4\|A\|_{\text{F}}^2), 1/(4\|A\|_{\text{op}})\}$, applied to both signs, gives the tail. The variance follows from $\mathbb{E}g_j^4 = 3$.

(c) Let g' be an independent copy, $g_\theta = g \cos \theta + g' \sin \theta$ and $\dot{g}_\theta = -g \sin \theta + g' \cos \theta$; for each θ the pair $(g_\theta, \dot{g}_\theta)$ is standard Gaussian. By the fundamental theorem of calculus and Jensen on $[0, \pi/2]$,

$$\mathbb{E}e^{s(F(g') - F(g))} \leq \frac{2}{\pi} \int_0^{\pi/2} \mathbb{E} \exp\left[\frac{\pi s}{2} \langle \nabla F(g_\theta), \dot{g}_\theta \rangle\right] d\theta \leq e^{\pi^2 s^2 \ell^2 / 8}.$$

Jensen in g' gives $\mathbb{E}e^{s(F(g) - \mathbb{E}F)} \leq e^{\pi^2 s^2 \ell^2 / 8}$; optimise in s .

(d) Let $\mathcal{N}_1, \mathcal{N}_2$ be $\frac{1}{2}$ -nets of the unit spheres of $\mathbb{R}^r$ and $\mathbb{R}^s$ with at most $5^r, 5^s$ points. For a vector w , choosing a net point y' near the unit vector $w/\|w\|$ gives $\langle w, y' \rangle \geq \frac{1}{2}\|w\|$; applying this twice gives $\|Z\|_{\text{op}} \leq 4 \max_{x \in \mathcal{N}_1, y \in \mathcal{N}_2} |x^T Z y|$. Each $x^T Z y$ is Gaussian with variance $\leq \sigma^2$; a union bound finishes. $\square$

Lemma A.2 (Squared Gaussian graphs). *Let Y be a centred symmetric Gaussian $n \times n$ matrix whose entries $(Y_{ij})_{i \leq j}$ have covariance $\preceq C_0 I/n$, and with $\max_i \mathbb{E} \sum_j Y_{ij}^2 \leq C_0 a$, where $d = an \geq 1$. For every $\zeta, \delta \in (0, 1)$ there is $C_{\zeta, \delta}$ (depending also on C_0) such that if $d \geq C_{\zeta, \delta} \log(2n)$ then $\|\mathcal{L}(Y) - \mathbb{E}\mathcal{L}(Y)\|_{\text{op}} \leq \zeta a$ with probability at least $1 - \delta$.*

Proof. $\mathcal{L}(Y) = \text{Diag}(\deg) - O(Y)$, where $\deg_i = \sum_{j \neq i} Y_{ij}^2$ and $O(Y)$ is the off-diagonal part of $Y \circ Y$. Each $\deg_i$ is a Gaussian quadratic form with coefficient matrix $A \succeq 0$, $\|A\|_{\text{op}} \leq C_0/n$, $\text{tr } A \leq C_0 a$, $\|A\|_{\text{F}}^2 \leq C_0^2 a/n$; by [Lemma A.1\(b\)](#) and a union bound, $\max_i |\deg_i - \mathbb{E} \deg_i| \leq \zeta a/2$ off probability $2ne^{-c\zeta d}$, and also $\mathbb{E} \max_i \deg_i \leq C(a + \log(2n)/n)$.

For the off-diagonal part let Y' be an independent copy. By Jensen, $\mathbb{E}\|O(Y) - \mathbb{E}O(Y)\|_{\text{op}} \leq \mathbb{E}\|O(Y) - O(Y')\|_{\text{op}}$, and $O(Y) - O(Y')$ is the off-diagonal part of $(Y - Y') \circ (Y + Y') = 2\tilde{Y} \circ \tilde{Y}'$, where $\tilde{Y} = (Y - Y')/\sqrt{2}$ and $\tilde{Y}' = (Y + Y')/\sqrt{2}$ are independent copies of Y . Conditionally on $\tilde{Y}$, the off-diagonal part of $\tilde{Y} \circ \tilde{Y}'$ is the Gaussian series $\sum_{i < j} \tilde{Y}'_{ij} \tilde{Y}_{ij} (E_{ij} + E_{ji})$, whose variance is, by [Lemma A.1\(a\)](#), $\preceq \frac{C_0}{n} \sum_{i < j} \tilde{Y}_{ij}^2 (E_{ii} + E_{jj}) \preceq \frac{C_0}{n} \max_i \deg_i(\tilde{Y}) I$. Hence

$$\mathbb{E}\|O(Y) - \mathbb{E}O(Y)\|_{\text{op}} \leq C \sqrt{\frac{\log 2n}{n}} \mathbb{E} \sqrt{\max_i \deg_i(\tilde{Y})} \leq Ca \left(\sqrt{\frac{\log 2n}{d}} + \frac{\log 2n}{d} \right),$$

and Markov's inequality finishes the proof. $\square$

B Relation to the original manuscript and its formalisation

The construction and almost all estimates are those of [\[5\]](#). The differences are as follows.

1. *Many-row seed for all small errors.* In [\[5\]](#) the many-row theorem assumes $\eta \leq c/d^2$, because the cubic remainder of the normalisation identity is bounded termwise by $O(t^3 d)$, and the operator norm of the noise by $t^2 \|Z\|_{\text{F}}^2 \leq 1$. [Lemma 4.4](#) uses the first-order constraint $X^T Z + Z^T X = 0$ a second time to centre the weights, giving $O(t^3)$, and the net bound controls $\|Z\|_{\text{op}}$. [Theorem 4.1](#) therefore holds for $\eta \leq c_*$.
2. *No block partition.* With 1., the global argument needs no random partition into blocks of size $\lceil Ld^6 \rceil$ (and no uniform-subset variance identity or averaging over block errors): the three regimes $d < D$, $n \geq Bd^2$, $n < Bd^2$ of [Proposition 6.1](#) cover everything, since $A \log(2Bd^2) \leq d$ for $d \geq D$.
3. *One toolbox.* The distance estimates, the potential, the static balancing theorem, the local correction lemma and the dense-core Poisson estimate are stated once ([Section 2](#)), and both seeds are routed through [Definition 2.13](#) and [Corollary 2.14](#).
4. *Complete sampling estimates.* The probabilistic statements used by the moderate seed are collected in [Lemma 5.7](#) and proved in full, including the variance bookkeeping behind the dense core.

Formal status. The Lean 4 development accompanying [\[5\]](#) proves [Theorems 1.1](#) and [1.2](#) by the original route (with the block partition and the $\eta \leq c/d^2$ many-row theorem). The argument of [Sections 4](#) and [6](#) here is not formalised; the other sections follow the formalised proof up to constants and presentation.

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